Algorithms, 2026 Fall, Homework 4
(Due: Oct 22)

October 12, 2026

Required Readings:

1. KT book, section 4.3 4.7 4.9

You can discuss homeworks with your classmates. But you will have to write down the solution on your own. You need to acknowledge other students who help you in the homework. If you read some other source that is helpful, you should list all of them.

Problems marked with (@) can be skipped by those who have extensive experience in OI (you need to provide a certificate to TA).
A footnote: Dynamic programming is a fairly standard algorithmic technique. But if you never see it before, it takes some effort to master it. Typically, the tricky part is to correctly set up the subproblems and this sometimes requires good intuition and some experience. If you never learn dynamic programming before, I highly suggest you to do more exercises to train yourself, especially you have a hard time finishing the above problems. (Both KT and CLRS books are very good sources of such exercises.)

Problem 1 : (@) Implement the weighted interval scheduling problem in $O(n \log n)$ time. You algorithm should return both the optimal value and the optimal solution (the subset of jobs). Write down the pseudo-code.

Problem 2 : (@) KT book pp.319 problem 8

Problem 3 : (@) KT book pp.320 problem 9

Problem 4 : (@) KT book pp.323 problem 13

Problem 5 : KT book pp.200 problem 23

Problem 6 : KT book pp.205 problem 32

Problem 7 : In this problem, we introduce a very useful and important concept in combinatorics, matroid. Matroids have very close connections with greedy algorithms and submodular functions (another important concept we will explore in later homeworks). The theory of matroids is a deep branch of combinatorics.

A finite matroid M is a pair $(E, \mathcal{I})$, where E is a finite set (called the "ground set") and $\mathcal{I}$ is a collection of subsets of E. Each element in $\mathcal{I}$ (which is a subset of E) is called an independent set (this notion has nothing to do with the notion of independent set in graph theory). Moreover, $(E, \mathcal{I})$ has to satisfy the following properties:

1. $\emptyset \in I$.
2. Every subset of an independent set is independent, i.e., if $A \subset B, B \in \mathcal{I}$, then $A \in \mathcal{I}$.
3. If A and B are two independent sets of $\mathcal{I}$ and A has more elements than B, then there exists an element $e \in A - B$ such that $B \cup \{e\}$ is an independent set.

Try to answer the following questions.

1. Suppose E is a set of real vector of the same length and

   $$\mathcal{I} = \{S \subseteq E \mid \text{All vectors in } S \text{ are linearly independent}\}.$$

   Prove that $(E, \mathcal{I})$ is a matroid.

2. Suppose E is the set of edges of a given graph G and

   $$\mathcal{I} = \{S \subseteq E \mid \text{All edges in } S \text{ constitute a forest of } G\}.$$

   (A forest is just a collection of trees. Note that a singleton node is also a tree) Prove that $(E, \mathcal{I})$ is a matroid.

3. Suppose each element $e \in E$ is associated with a weight $c_e \geq 0$. Design a polynomial time algorithm to find a max-weight independent set. In other words, the algorithm should output an independent set $A \in \mathcal{I}$ such that $\sum_{e \in A} c_e$ is maximized. You need to prove the correctness of your algorithm. (Hint: Kruskal's algorithm.)

(Sidenote: There are several important problems which also fits to the framework of matroid. We will cover some of them in the later part of this course. Matroid also has a deep connection with linear programming and other set systems (like polymatroid and greedoid). You will learn some of them later.)